

Functional and differential equations

— 5

Second-order differential inequalities and estimates of
solutions

Homework

13 сентября 2026 года

Theoretical minimum. A linear differential inequality with constant coefficients is integrated by factoring the operator into first-order factors and applying the integrating factor twice; the outcome is an explicit comparison with the solution of the corresponding equation. The Riccati substitution converts a second-order equation into a first-order inequality, and the length of an interval on which the Riccati variable runs from $+\infty$ to $-\infty$ is bounded below — this is the same statement as Sturm's comparison theorem in disguise. The maximum principle says that a function with $y'' \geq 0$ has no interior maximum, and it immediately yields uniqueness for boundary value problems. For integral quantities the right tools are Wirtinger, Poincaré and Opial. Finally, Lyapunov's inequality bounds the potential from below when a solution has two zeros, and is the sharp quantitative form of the oscillation theorems.

1. Prove that $f'' \geq f$ on $[0, \infty)$ with $f(0) = f'(0) = 0$ implies $f \geq 0$, and that $f'' \leq f$ with the same data implies $f \leq 0$; then find the sharp constant c for which $f'' \geq cf$ forces f to be non-negative.
2. Let f be twice differentiable with $f + f'' \geq 0$ on $[0, \pi]$ and $f(0) = f(\pi) = 0$. Prove that $f \geq 0$ there, and deduce $\int_0^\pi f(x) \sin x \, dx \geq 0$ (compare with problems 8 and 10 in the homework of Differentiable functions – 6).

3. Prove the Bihari–LaSalle inequality in the second-order form: if $u'' \leq a(x)g(u)$ with positive non-decreasing g , the solution is bounded by the solution of the corresponding equation as long as the latter exists.
4. Prove the Fite–Wintner theorem: if $q \geq 0$ is continuous on $[0, \infty)$ and $\int_0^\infty q(x) dx = +\infty$, then every solution of $y'' + q(x)y = 0$ has infinitely many zeros; and show that $\int_0^\infty q < \infty$ does not by itself imply non-oscillation.
5. Prove that a function with $f''' \geq 0$, $f(0) = f'(0) = f''(0) = 0$ is non-negative, convex and increasing, and generalise to an arbitrary linear operator with non-negative characteristic roots.
6. Let $u \geq 0$ satisfy $u'' \geq u^2$ on $[0, \infty)$ with $u(0) > 0$, $u'(0) > 0$. Prove that u blows up in finite time and estimate the blow-up time from below and above.
7. Prove that if $y'' + q(x)y = 0$ has a positive solution on (a, b) , then no solution has two zeros there, and deduce the criterion for non-oscillation in terms of a Riccati inequality.
8. Let f be convex on $[0, 1]$ with $f(0) = f(1) = 0$. Prove that $\int_0^1 |f| \leq \frac{1}{8} \max |f''|$ when $f \in C^2$, and find the extremal function.
9. Prove the discrete analogue of the maximum principle: if $u_{n+1} - 2u_n + u_{n-1} \geq 0$ for $1 \leq n \leq N - 1$, then $\max u_n$ is attained at $n = 0$ or $n = N$; deduce uniqueness for the discrete boundary value problem and compare with the continuous case.
10. Let y solve $y'' + q(x)y = 0$ with q continuous and $\int_0^\infty x|q(x)|dx < \infty$. Prove that every solution is asymptotically linear, that is $y(x) = \alpha x + \beta + o(1)$, and identify the map from initial data to (α, β) .